

EECS 4314 E - Advanced Software Engineering
Project Proposal - Group 10

Q1. Name of your product: Savr

Q2. Briefly describe the core concept of your application in terms of user functionality.

Savr is a dynamic web-based social media application focused on recipe sharing and food discovery. Our platform allows users to create accounts, securely log in, and interact with a community centered around cooking and recipes. Users will be able to upload recipes that include ingredients, preparation instructions, images, and optional tags or categories. Other users can browse recipes, leave comments, provide ratings, and save recipes for future reference. The application will also support social features such as following other users to receive updates whenever new recipes are uploaded. To improve usability and user experience, recipes can be sorted and filtered based on criteria such as highest rated, most viewed, newest uploads, or trending recipes. Our goal is to create an organized and interactive environment where users can easily discover new recipes and engage with other members of the community. Savr combines recipe management with social media functionality in a responsive and user-friendly web application.

Q3. List the names of each person in the project group.

- Manjot Kaur
- Gurnoor Kahlon
- Nathan Kwok
- Andriy Lytvyn
- Larissa Singh
- Rishab Yadav

Q4. List programming languages and web development frameworks that you know or want to use.

Frontend: React, JavaScript, HTML/CSS

Backend: Java (Springboot)

Database: MySQL

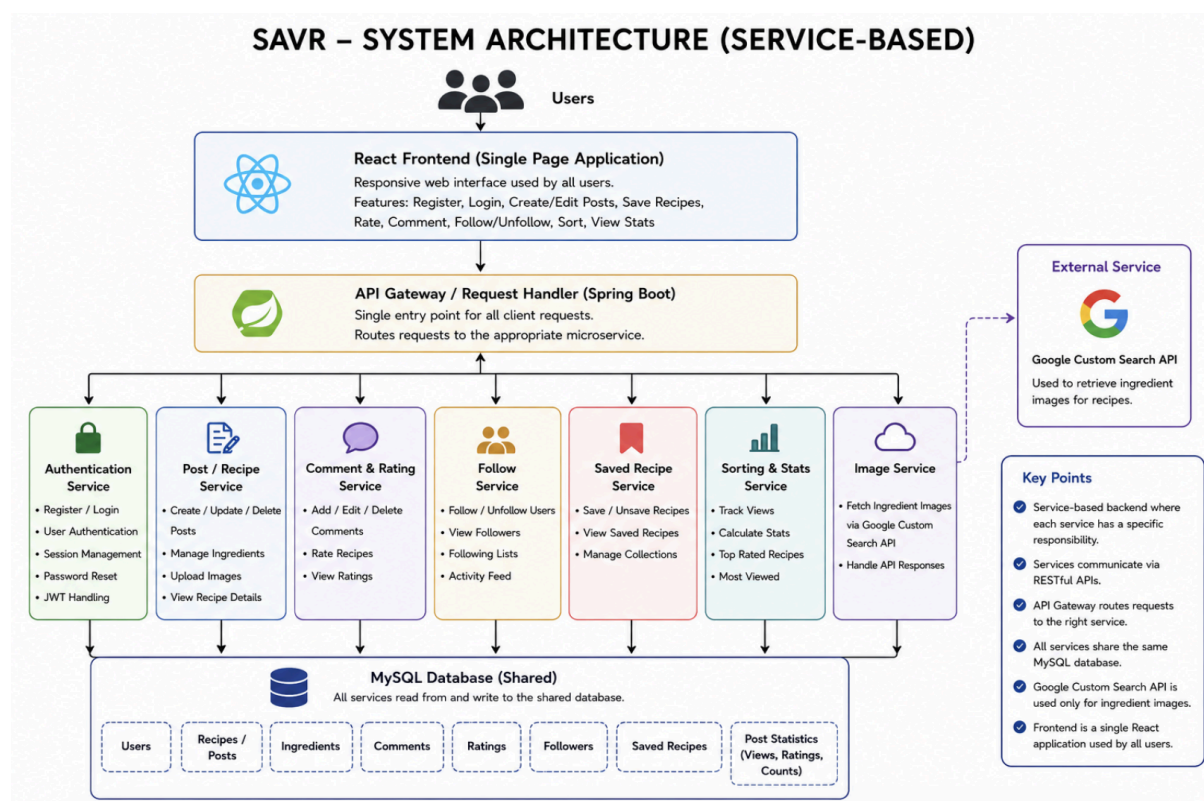
Q5. Provide a rough diagram or description of your system architecture, which project requirements you strive to implement, and how this fulfils the learning goals for this course.

Savr will follow a service-based architecture style with a single monolithic user interface. The system will consist of a React frontend connected to backend services developed using Spring Boot, with MySQL used for persistent data storage. The frontend layer will provide a single user interface where users can interact with the application by creating accounts,

logging in, uploading recipes, viewing recipes, commenting, rating posts, and following other users. The interface will also allow users to sort recipes by highest rating, most viewed, newest uploads within the last week, or all-time popularity. The backend service layer will handle the main application logic and communication with the database. Different services within the backend will manage:

- User registration and authentication
- Recipe posting, editing, saving, and viewing
- Recipe rating and comment management
- Following and unfollowing users
- Tracking post statistics such as views and ratings
- Managing visibility of posts depending on follow status
- Sorting and filtering recipes
- Retrieving ingredient-related images through an external Google API

The MySQL database layer will store information related to user accounts, recipes, comments, ratings, followers, and saved recipes.



Through the development of Savr, our team will gain practical experience applying the software engineering concepts and learning outcomes covered throughout the course.

- Have a deep understanding and practical knowledge of software architecture models and documentation used to design large-scale software systems.
 - The project directly addresses this by giving us experience designing the architecture of a full-stack web application.

- We will also create system documentation and architectural UML diagrams throughout development.
- Have a deep understanding and practical knowledge of non-functional requirements and how to identify and include them in the design of a software system.
 - We will consider requirements such as maintainability, scalability, usability, and modularity while designing the application.
 - A README file will also be included in the GitHub repository explaining how to install, run, and use the website.
- Know various software architectural styles and be able to select a style fitting for a particular problem.
 - Multiple architectural styles were considered before selecting a service-based architecture style for Savr.
 - The selected architecture will continue to be evaluated and refined throughout development.
- Know common software patterns and demonstrate their usage within the course project.
 - We plan to learn and apply software design patterns where appropriate to improve code organization and maintainability.
 - These patterns will help support modular and reusable development practices.
- Know software reuse techniques and demonstrate this knowledge through the course project.
 - Reusable components and shared functionality will be implemented throughout both the frontend and backend of the application.
 - This will help reduce redundancy and improve maintainability.
- Use industry-standard development tools, including source control and continuous integration practices.
 - Technologies such as React, Spring Boot, and MySQL were selected because they are widely used in industry.
 - GitHub and GitHub Desktop will be used for source control, collaboration, and version control throughout the project.
- Use architectural evaluation techniques and tools to evaluate the architecture of a software system.
 - Our team evaluated multiple possible architectures before selecting the current system design.
 - The architecture will continue to be reviewed as additional project requirements are implemented.
- Practice project management, source control, and team-based development experience while working on a large-scale software project.
 - We will hold regular meetings to discuss progress, assign tasks, and manage development timelines.
 - Team collaboration and version control workflows will be used throughout the project to support coordinated development.
- Develop models of software systems and express them using documentation and diagrams such as UML.

- UML diagrams and architectural representations will be created to model the structure and functionality of the system. These models will help improve communication, planning, and system organization during development.
- Identify risks associated with a software project and develop plans to manage them.
 - Potential risks such as integration challenges, time management, and feature complexity will be discussed throughout development.
 - Responsibilities will be divided among team members to help manage workload and reduce development risks.